

VISHAL SELVAN

Electronics and Communication Engineer

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PROFILE

Final year Electronics and Communication Engineering student with a strong interest in VLSI Design, Embedded Systems, and Avionics. Experienced in developing hardware-based projects with knowledge of ASIC Design, SoC Architecture, IP Design, PCB Design, Digital Electronics and Circuit Analysis. Passionate about learning new technologies and seeking opportunities to apply technical knowledge and contribute to innovative engineering solutions.

EDUCATION

B.TECH - Electronics And Communication Engineering 2023– 2027
(Cyber Physical Systems)
SASTRA DEEMED UNIVERSITY, Thanjavur | CGPA-7.11

GRADE 12 2022– 2023
ZION MATRICULATION HIGHER SECONDARY SCHOOL, Chennai | PERCENTAGE - 89.6

TECHNICAL SKILLS

EDA (Schematic & PCB)
(Tools - KiCAD, Altium, Proteus, Cadence)
Programming - C, Python, Embedded C
Hardware Description Languages - Verilog, Systemverilog
(Tools - Xilinx Vivado, Gowin IDE, ModelSim, Quartus Prime)
Protocols - CAN, I2C, UART/USART, SPI
Embedded Systems - STM32, ESP32, PLC(ladder logic), ARM Cortex-M, FreeRTOS, Linux
(Tools - Stm32cubeIDE, ArduinoIDE, Edge Impulse, Open PLC, ThingSpeak)
Architecture - RISC-V, Pipeline Design, ISA, RTL Design
Simulation & Analysis
(Tools - MATLAB & Simulink)

PROJECTS

32 Bit RISC-V Processor : Designed and developed a 32-bit RISC processor architecture using Verilog HDL with custom instruction execution and control logic. Implemented GPIO interfacing and integrated peripheral modules for hardware communication and testing. Validated the processor functionality through FPGA-based simulation and real-time hardware verification.

BeagleBone Black Rev C : Recreated the schematic and 6-layer PCB layout of the BeagleBone Black Rev C using KiCad. Applied PCB design principles including component placement, signal routing, power and ground plane planning, differential pair routing, and high-speed PCB techniques. Verified the design using Electrical Rules Check (ERC) and Design Rules Check (DRC).

CAN Controller Implementation On FPGA : Developed and implemented a CAN controller on an FPGA using Verilog HDL for real-time data communication. Implemented CAN transmission and reception logic, interfaced with a CAN transceiver, and verified interoperability through hardware testing with an STM32 microcontroller.

VGA Controller on FPGA : Designed and developed a VGA controller in Verilog HDL on the Arty A7 FPGA to generate 640×480 @ 60 Hz video output. Implemented object rendering, collision detection, score tracking, and user-controlled movement for an interactive game. Verified the design using Vivado and Integrated Logic Analyzer (ILA).

Line Following Robot : Designed and developed a line following robot without using a microcontroller by implementing hardware based control logic. Utilized IR sensors and an LM358 comparator for path detection and signal processing. Integrated motor driver circuitry for real-time autonomous line tracking and movement control

ORGANIZATION

CoreSelva — Educational Hardware IP Platform

Co-Developer, Contributed to CoreSelva, an educational hardware IP platform focused on RISC-V and FPGA-based projects. assisted in showcasing embedded systems, processor design, and hardware experimentation projects through technical documentation, GitHub integration, and web-based project presentation.

INTERNSHIPS

DELLNER INDIA PVT LTD

Gained industrial exposure by observing PLC-controlled automation systems used in the bending, welding, and fabrication of steel rods for train gangway assemblies. Developed an understanding of industrial automation workflows, PLC-based machine control, manufacturing processes, production line operations, quality control procedures, and workplace safety practices in a real-time industrial environment.

CSSIR-CENTRAL ELECTRONICS ENGINEERING RESEARCH INSTITUTE

Gained hands-on project experience by designing and testing an Automatic Fan Speed Controller to adjust fan speed automatically based on temperature changes. Enhanced practical knowledge of embedded systems, sensor interfacing, relay control, and hardware integration through implementation and testing. Improved understanding of real-time monitoring and system-level debugging during hardware development.